Project Overview

Project Title : *AlgoNova* Team Members

- **Team leader :** Seddiki Mohamed : Backend Developer , System Architect and database manegment
- **Frontend Developers :**Alioueche Ahmed and Haddad Hadjer
- **UI/UX Designer :** Mecheri Radhia
- **Documentation Manager & Content :** Abdelkebir Raouane

Project Description

The **AlgoNova** platform is an educational environment designed to teach algorithms through interactive and theoretical lessons, combined with practical exercises using a custom educational programming language. It integrates a web interface, backend API, centralized database management via Supabase, and a desktop application. The system features user authentication, role-based access control (Admin and Regular User), and an admin dashboard to manage lessons and users. Additionally, it includes an AI-powered chat assistant providing real-time explanations, debugging support, and step-by-step guidance.

AlgoNova aims to create an intelligent, structured, and engaging learning environment that enhances algorithm understanding and student interaction.

Problem Statement

Many students face difficulties understanding algorithms due to the abstract nature of concepts, limited personalized support, and lack of real-time feedback during problem solving. Traditional learning platforms often provide static content without intelligent interaction, making it difficult for students to identify, analyze, and correct their mistakes. Therefore, there is a need for an integrated system that combines structured learning, practical execution, and AI-powered assistance to provide immediate guidance and enhance the overall learning experience.

Project Objectives

The main objectives of this project are:

1. Develop a complete educational platform for algorithm learning.
2. Design and implement a custom algorithm execution engine (MyAlgo language).
3. Implement secure authentication and role-based access control.
4. Create an admin dashboard for lesson and user management.
5. Integrate an AI chat assistant for interactive support and code explanation.
6. Develop a desktop application version of the platform.
7. Ensure smooth frontend-backend-database integration.
8. Deliver a scalable, maintainable, and user-friendly system.

Project Planning

Project Timeline

Phase	Tasks	Deadline
1-Analysis & Design	Requirement analysis, system architecture design, UI/UX wireframes	Week 1–2
2-Core Development	MyAlgo engine development, backend API setup	Week 3–5
3-Database & Authentication	Supabase setup, authentication system implementation	Week 5–6
4-Frontend Development	User interface, admin dashboard, AI Chat interface	Week 6–8
5-AI Integration	AI API integration, testing and optimization	Week 8–9
6-Integration & Packaging	Frontend-backend integration, Electron packaging	Week 10
7-Testing & Documentation	System testing, debugging, final documentation	Week 11–12

Task Distribution

- **Seddiki Mohamed (Team Leader – Backend) :**Backend API development , AI integration logic, frontend-backend integration, MyAlgo language development, Supabase integration, authentication system, database management , GitHub code integration control, final system integration.
- **Alioueche Ahmed & Haddad Hadjer (Frontend) :**React pages, Ul components, lesson display,login/register interface (UI), admin dashboard interface.
- **Mecheri Radhia (UI/UX) :**Interface design, user experience optimization, layout and interaction design.
- **Abdelkebir Raouane (Content & Documentation) :**Project documentation, final report preparation , Project documentation, algorithm lessons preparation, lesson file collection.

Milestones and Deliverables

- **M1 (Week 2):** Approved system design and UI prototype.
- **M2 (Week 5):** Functional backend API and MyAlgo engine.
- **M3 (Week 8):** Complete frontend interface and dashboard.
- **M4 (Week 9):** Fully integrated AI Chat assistant.
- **M5 (Week 10):** Desktop application package ready.
- **M6 (Week 12):** Final tested system and complete documentation submitted.

Technical Requirements

Programming Languages : C++, C#, typescript ,python
Frameworks & Technologies: • React , vite (Frontend) • ASP.NET Core (Backend)
• Electron (Desktop Application) • Git & GitHub (Version Control)
• Supabase (Authentication & Database) •Docker
Database: PostgreSQL (via Supabase)

Development Tools: Visual Studio, VS Code, Postman